

Subhashini Venkatachalam | Gen Ai Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 4 years of experience developing backend services, cloud-native applications, microservices, and AI-enabled solutions using Python, Java, C#, Spring Boot, .NET, Azure OpenAI, and LangChain. Experienced in building REST APIs, integrating generative AI and retrieval-augmented generation (RAG) workflows, developing machine learning solutions, and implementing event-driven architectures across AWS and Azure environments. Skilled in database engineering, CI/CD automation, and production application monitoring, with a focus on scalable and maintainable software systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, C#, SQL, JavaScript, TypeScript

AI & Machine Learning: Azure OpenAI, LangChain, Retrieval-Augmented Generation (RAG), TensorFlow, Keras, ResNet-50, Convolutional Neural Networks (CNN), Deep Learning, Image Classification, Scikit-learn, KNIME

Backend & APIs: Spring Boot, .NET Core 6, ASP.NET MVC, Node.js, REST APIs, GraphQL, Microservices, Event-Driven Architecture

Frontend: Angular 15, React.js, Redux, HTML5, CSS3, Tailwind CSS

Databases & Data Engineering: PostgreSQL, SQL Server, Snowflake, DynamoDB, MongoDB, Redis, Databricks, SSIS, ETL Pipelines

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, GitHub Actions, Jenkins, Azure DevOps, CI/CD

Testing & Monitoring: JUnit, Mockito, NUnit, Selenium, Postman, SonarQube, Datadog, Amazon CloudWatch, Power BI

PROFESSIONAL EXPERIENCE

Wayfair, USA | Gen AI Software Engineer

Jun 2025 – Present

- Optimized PostgreSQL and DynamoDB query execution for inventory data workflows, reducing average retrieval latency by 32% through query tuning, indexing, and targeted access-pattern improvements.
- Implemented Kafka Streams and Redis caching for streaming inventory workflows, improving repeat-read response times by 28% and reducing redundant database lookups across AI-enabled application services.
- Reduced machine learning model release time from 60 minutes to 18 minutes by automating CI/CD workflows with GitHub Actions, Jenkins, and Kubernetes.
- Streamlined asynchronous AI inference processing by connecting AWS Lambda producers with API Gateway, SNS, and SQS workflows, reducing average downstream processing time by 35%.
- Developed Java and Spring Boot microservices integrating LangChain capabilities across 4 inventory and data-synchronization services, improving modularity and deployment independence for enterprise workflows.
- Improved production observability across distributed AI applications by configuring Datadog and CloudWatch monitoring, reducing average troubleshooting time by 25% through centralized service and infrastructure telemetry.

Cognizant, India | Software Engineer

Oct 2021 – Jul 2024

- Optimized Snowflake analytical workloads through partitioning and clustering strategies, reducing average dashboard query execution time by 30% across high-volume business intelligence workloads.
- Engineered SQL Server, SSIS, and Databricks ETL pipelines processing more than 1 million healthcare records weekly, reducing recurring batch-processing time by 25% through automated retries and pipeline tuning.
- Automated software release workflows in Azure DevOps, reducing manual release time from approximately 2 hours to under 30 minutes through automated build and deployment processes.
- Integrated an internal generative AI chatbot using Azure OpenAI, Python, and LangChain, reducing average employee knowledge-search time by 35% through retrieval-augmented generation workflows.
- Modernized legacy monolithic applications into modular .NET Core microservices across 5 core services, reducing scheduled deployment downtime by 30% and enabling more independent release cycles.
- Engineered C#/.NET and Java REST microservices with token-based authentication, integrating AWS SNS producers and SQS consumers to support asynchronous communication across 6 backend service workflows.

PROJECTS

Medicinal Plant Image Recognition (TensorFlow & Keras)

- Developed a ResNet-50 CNN classifier in Python utilizing transfer learning and image augmentation to identify 34 medicinal plant species from 1,550 images.
- Achieved a 0.98 F1-score on a held-out test set, proving model reliability for automated computer vision identification under varied lighting conditions.

Elephant Call Acoustic Classification (KNIME)

- Built a machine learning pipeline using 886 acoustic samples to classify wildlife vocalizations.
- Evaluated Decision Tree, Neural Network, and Logistic Regression models, achieving a peak accuracy of 75.6% with the Decision Tree.

EDUCATION

Master of Computer Information Systems, Colorado State University, Fort Collins, CO

Jan 2024 – May 2025

CERTIFICATION

- Microsoft Certified: Azure Fundamentals (AZ-900)**